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A new approach of DGA interpretation technique for transformer fault diagnosis

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ABSTRACT

Dissolved Gas Analysis (DGA) is one of the most common techniques to detect the incipient faults in the oil-filled power transformers. In this paper, a new approach of DGA technique is proposed to overcome the conflict that takes place in the traditional interpretation techniques for transformer fault diagnosis. The new approach is based on the analysis of 386 dissolved gas samples data set that collected from the Egyptian electric utility chemical laboratory as well as from credited literatures. These data sets are used to build the technique model and also as a tested data set to get the technique's accuracy. The new approach DGA diagnoses the transformer fault types based on the gas concentration percentage limit of the sum of main five gases (Hydrogen (H_2), Methane (CH_4), Ethan (C_2H_6), Ethylene (C_2H_4), and Acetylene (C_2H_2)) and some suggested gases ratios depending on the sample data set analysis. The validation of the proposed approach of DGA technique is satisfied by comparing its results with the results of the IEC Standard Code, Duval triangle and Rogers methods for the collected data set. The results refer to the ability and reliability of the new approach in transformer faults diagnostic.

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Introduction

Power transformers are vital, important and expensive components in electric power systems. The discontinuity of the power transformers is uneconomical and causes a great financial loss for the electrical utilities. Some factors affect on the transformer condition and lead to an increase in loading power transformer such as its ageing, rising energy consumption and liberation. Therefore, an increase in thermal, electrical and mechanical stresses of transformers will be developed. Hence, it is very important to have a suitable and effective evaluation of early detection of the incipient faults of power transformer conditions. Dissolved Gas Analysis (DGA) in oil is a widespread method that is used to identify the incipient faults in oil-filled power transformers. As a result of the electrical and thermal stresses, insulating transformer oil maybe decayed and several gases are released and dissolved in oil. These gases include hydrogen (H_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), acetylene (C_2H_2), carbon monoxide (CO) and carbon dioxide (CO_2) in which their concentrations are expressed in part per million (ppm).

The most criteria that are commonly used for dissolved gas analysis to interpret the incipient transformer faults are Dornenburg method, Rogers' method, key gas method, Duval Triangle method, IEC standard Code [1–3]. The above-mentioned criteria use some gas ratios for fault diagnosis, and some of these methods compare gas concentrations to the specified levels to evaluate a transformer's condition. All these conventional methods are simple and easy to implement, but their diagnostic accuracy is still limited and very sensitive to uncertainties in DGA data. However, some problems according, to Duval Triangle method are due to detection failure with the partial discharge (PD) fault and furthermore some interference between thermal and electrical faults is found like in the area of electrical and thermal fault (DT). In IEC technique, the interference between low discharge energy (D1) and high discharge energy (D2) can lead to the mislead results [1]. The results obtained from Rogers's method are not concise in case of detecting all faults except the low thermal (T1) fault. In addition, some recent DGA techniques can't to separate between the low and medium thermal faults (T1 and T2) as in [4]. Generally, new gas diagnosis features are to be attained in order to improve the transformer condition monitoring [4].

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Table 1
The number of samples for each fault type.

Fault type	PD	D1	D2	T1	T2	T3	Overall
Lab samples	27	42	55	70	18	28	240
Literatures samples	16	27	60	11	6	26	146
Total	43	69	115	81	24	54	386

Table 2
Matching between IEEE std. and laboratory measurement of Key gases.

Fault type	PD	D1	D2	T3
IEEE standard	H_2	H_2	H_2+C_2H_2	C_2H_4
Laboratory	H_2	H_2	C_2H_2	C_2H_4

Nowadays, more efforts are exerted to build computational models to improve the traditional interpretation techniques using artificial Intelligence (AI) to analyze incipient fault in transformers. These AI methods are: Artificial Neural Network (ANN) [5,6], Fuzzy Logic [7,8], Support Vector Machine (SVM) [9,10], Expert System [11,12], Hybrid System [13,14] and Graphical Techniques [15,16].

In this paper, a new approach of the DGA is developed. It is based on both the concentration percentage of the dissolved gases to the sum of the main five gases and some gas ratios. The five main dissolved gases are Hydrogen (H_2), Methane (CH_4), Ethane (C_2H_6), Ethylene (C_2H_4), and Acetylene (C_2H_2). Each gas concentration is calculated as its percentage with respect to the total sum of the five selected gases. Six main faults are concerned, PD, D1, D2, T1, T2 and high thermal fault (T3). Each fault type is concluded, based on the limit of each gas concentration percentage and some of the proposed gas ratios. 386 samples are used to develop the new approach technique that 240 dissolved gas of the oil samples from the chemical laboratory of the electric utility in Egypt and the others are from literatures. A validation of the proposed approach technique is satisfied by comparing its performance with the most famous methods that are IEC Code, Duval Triangle, and Rogers' four ratios for these dissolved gases of the oil samples. The comparison refers to the ability of the new approach technique to diagnose the transformer faults according rather than the other methods.

Proposed DGA interpretation technique

Till now, the current interpretation techniques didn't give correctly accepted results for transformer fault detection; therefore, it is an open area for proposed idea to overcome the problems of those DGA techniques. The accuracy limit of the traditional DGA techniques is still a great issue to diagnose the transformer faults due to the electrical and thermal stresses. The proposed rules of the new approach DGA technique are developed based on data set samples from laboratory (electrical utilities in Egypt) and credited literatures. These rules depend on the concentration limit of each gas, according to each transformer fault type with some gas ratios to separate the interference between the fault types. The number of samples according to the fault type is explained in Table 1.

The total number of actual DGA data is 386 data samples. 240 data samples are collected from practical cases into the Egyptian Electricity Network. These practical cases were analyzed by advanced software that supplied with the DGA analyzer which used by the utility. This software is considered accurate and

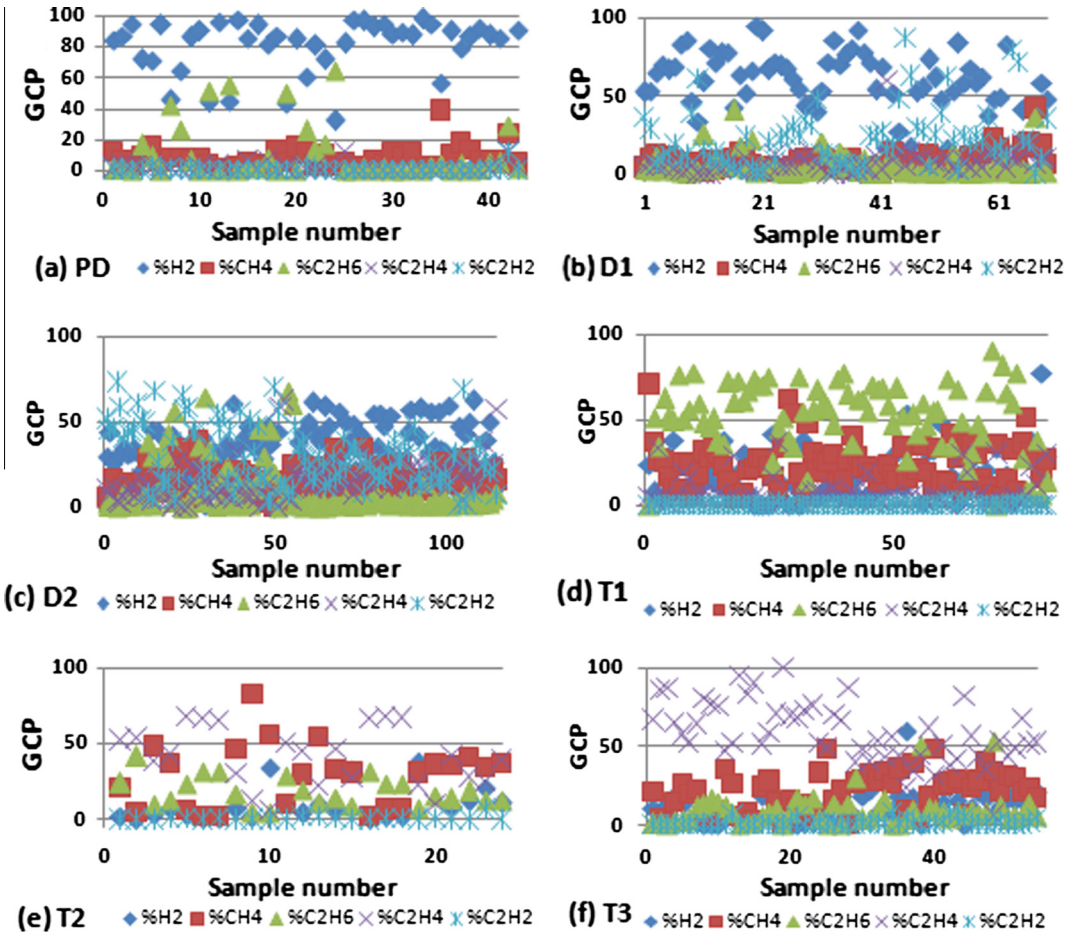


Fig. 1. Gas concentration percentage (*GCP*) for each fault type.

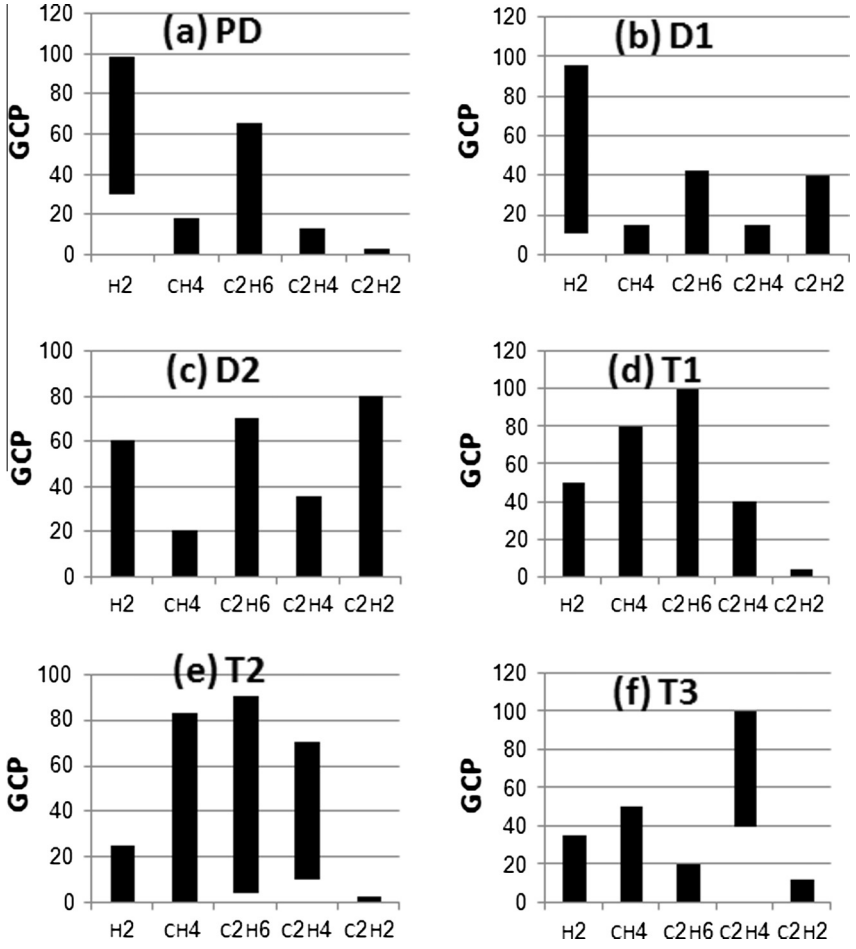


Fig. 2. Gas concentration percentage (*GCP*) limit of each gas according to fault type according to (2).

Table 3
Gas concentration limits.

Fault type	H_2%	CH_4%	C_2H_6%	C_2H_4%	C_2H_2%
PD	30–98	⩽18	⩽66	⩽13	⩽2.5
D1	10–96	⩽14.5	⩽42	⩽5	⩽40
D2	⩽61	⩽40	⩽70	⩽35	⩽80
T1	⩽50	⩽80	⩽100	⩽40	⩽4
T2	⩽25	⩽83	4–90	10–70	⩽2
T3	⩽35	⩽50	⩽20	30–100	⩽12

reliable, especially that, when some transformers were inspected, the fault type matched with the software diagnosis. This inspection was done based on the IEEE Std. C57.104-2008, in section 6.6 that is entitled "Evaluation of Possible Fault Type by the Key Gas Method". The second row in following Table 2 shows the standard IEEE C57.104-2008 based key gases that indicate the fault type. When the measured data are evaluated, the third row is obtained where there is a great matching between the standard and the laboratory measurements. Accordingly, the reliability of the laboratory data is confirmed. In addition, the considered data for evaluating the proposed technique did not rely on these data only, where other 146 data samples from published cases were used. The feature extraction proposed in this section, proposed algorithm procedure, and its evaluation is as follows.

Feature extraction

The New approach DGA technique is based on determination of the gases' concentration percentage that expresses the transformer fault types. There are most common six fault types that can occur in the transformer oil. They are PD, D1, D2, T1, T2, and T3. If the technique fails to interpret the type of the fault, then the undetermined fault (UD) case is developed. The procedure considered finding the gases-based fault features as follows. First, the data of the dissolved gas are collected. Second, the sum of five created gases during the fault is Hydrogen (H_2), Methane (CH_4), Ethane (C_2H_6), Ethylene (C_2H_4), and Acetylene (C_2H_2) is calculated. This summation is called Total Combustion Gases (*TCG*) as in:

$$TCG = H_2 + CH_4 + C_2H_6 + C_2H_4 + C_2H_2 \quad (1)$$

Therefore, each of gas percentage is evaluated using the following vector form gas concentration percentage (*GCP*):

$$GCP = \left(\frac{H_2}{TCG} \quad \frac{CH_4}{TCG} \quad \frac{C_2H_6}{TCG} \quad \frac{C_2H_4}{TCG} \quad \frac{C_2H_2}{TCG}\right)^T \times 100\% \quad (2)$$

The rules of each transformer fault type are extracted using *GCP* distributions according to the sample numbers as in Fig. 1. The horizontal axis and secondary vertical axis illustrate the number of samples according to each fault type for all five main gases, i.e., 43 samples in PD case and 115 samples in D2 case. Fig. 2 shows the percentage of gas concentration limit of the dissolved gases, according to each fault type. The gas concentration limits are proposed based on the empirical study of the data set samples. Fig. 2a confirms that H_2% is the most generated gas due to PD fault type where it is in the range 30–98%. Further gas due to this fault is the C_2H_6 with percentage ranges less than 66%. The gas evaluation due to D1 fault type is shown in Fig. 2b where H_2 varies from 10%

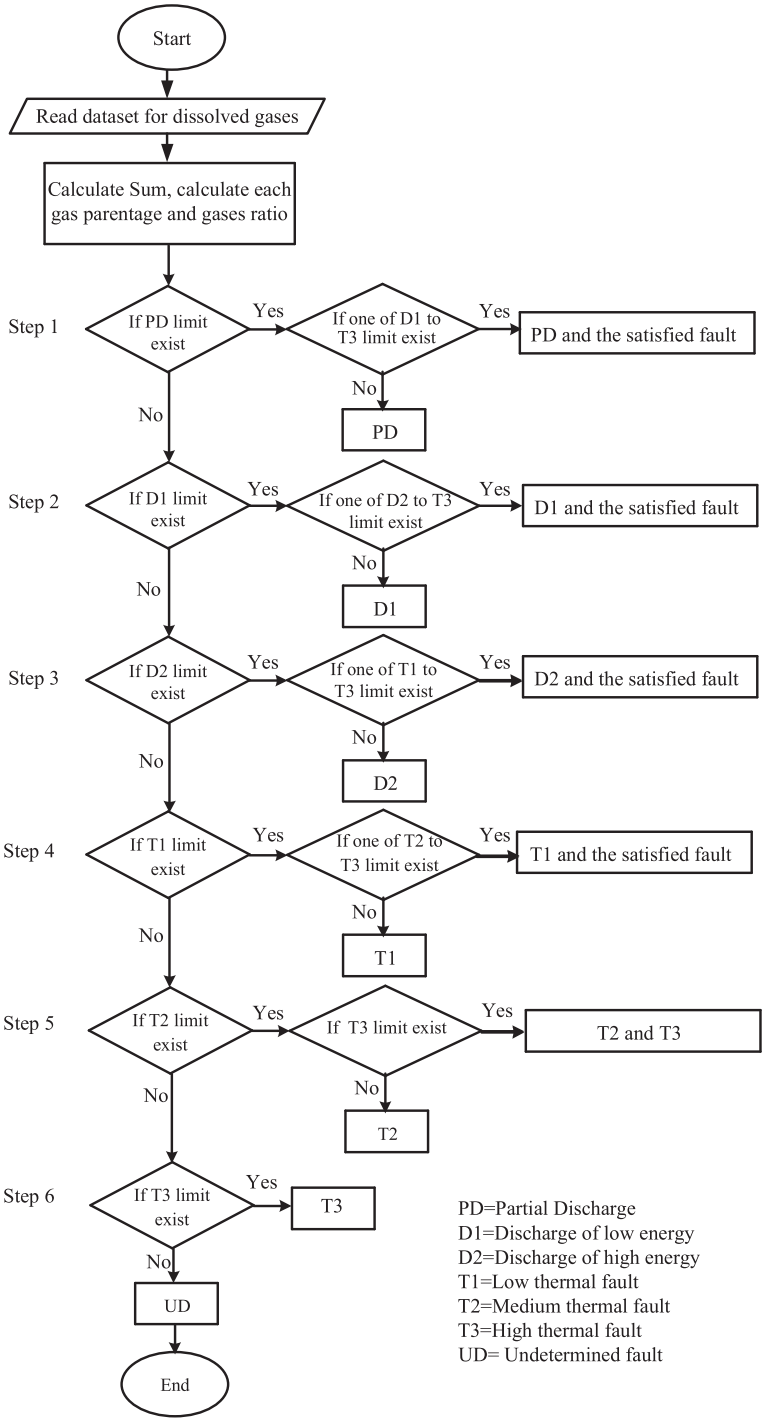


Fig. 3. Flowchart of the new approach DGA technique.

Table 4
Agreement accuracy percentage between new approach DGA and common DGA techniques for 386 samples according to fault types.

Fault type	Sample	New approach DGA	IEC	Duval	Rogers
PD	43	56.97	37.21	48.83	23.26
D1	69	60.14	36.23	72.46	1.45
D2	115	88.69	44.34	59.13	56.52
T1	81	73.23	60.49	48.14	90.12
T2	24	64.58	66.67	29.16	12.50
T3	54	75	68.52	92.59	38.89
Overall	386	71.5	50.26	60.88	44.82

Table 5
Agreement accuracy percentage between New approach DGA and common DGA Techniques for 386 samples according to source of dataset.

Ref.	Samples	New approach DGA	IEC	Duval	Rogers
Lab	240	75.38	41.25	48.75	37.92
[10]	8	43.75	25	50	25
[15]	5	70	40	100	40
[17]	99	68.18	71.72	93.94	58.59
[18]	19	55.26	84.21	57.9	68.42
[19]	6	58.33	50	50	16.67
[20]	9	61.11	33.33	44.44	66.67
Overall	386	71.5	50.26	60.88	44.82

Table 6
The overlap that take places between fault types for some dataset.

Ref.	Gas concentration (ppm)					Actual	Diagnose	AF
	H_2	CH_4	C_2H_6	C_2H_4	C_2H_2			
Lab	117	17	1	3	1	PD	PD, D1	0.5
[17]	32,930	2397	157	0.001	0.001	PD	PD, D1	0.5
[17]	78	20	11	13	28	D1	D1, D2	0.5
[17]	1230	163	27	233	692	D1	D1, D2	0.5
[17]	8200	3790	250	4620	277	D2	D2, T1	0.5
[17]	13	3	1	3	6	D2	D1, D2	0.5
[15]	130	140	2	120	0.001	T1	D2, T1	0.5
[15]	78	66	283	2.6	0.001	T1	D2, T1	0.5
[10]	30.4	117	44.2	138	0.1	T2	T1, T2	0.5
[18]	27	90	42	63	0.2	T2	T1, T2	0.5
[17]	1100	1600	221	2010	26	T3	T2, T3	0.5
[17]	290	966	299	1810	57	T3	T2, T3	0.5

to 96%, while C_2H_6 is reduced to the range less than 42%. Further created gas due to D1 fault is C_2H_2 that is less than 40%. Fig. 2c indicates that the D2 fault produces gases' C_2H_2, C_2H_6 and H_2 that are less than 80%, 70% and 61% respectively. The fault T1 generates C_2H_6 over the whole range up to 100%, CH_4 and H_2 are less than 80%, and 50% respectively as in Fig. 2d. For the fault type T2 shown in Fig. 2e, it produces gas C_2H_6 varying from 4% to 90%, CH_4 less than 83%, C_2H_4 varying from 10% to 70%. Fig. 2f indicates that the T3 fault creates C_2H_4 in the range 30–100%, CH_4 and H_2 are less than 50% and 35% respectively. Such gas concentrations based on the fault type are summarized in Table 3.

From the above discussion, each fault type has a distinct gas, such as Hydrogen H_2 for PD fault. Hydrogen (H_2) and Ethan (C_2H_6) and Acetylene (C_2H_2) are the key gas in case of D1. Only Acetylene (C_2H_2) is the dominant gas in case of high discharge energy. The Ethan (C_2H_6) refers to T1 but in case of T2 and T3 thermal faults; the Ethylene (C_2H_4) considers the distinct gas. Concerning such information, an algorithm to determine the fault is presented as in the following subsection.

The pervious gas limits are derived from the collected data set directly and these limits are concluded, based on the concentration of the dissolved gases according to each fault type. Maybe these limits can change when the collected data set change, but this change is not significant based on the key gas theory of each fault type. As in Fig. 1, the key gas for the PD fault is H_2 and when the data is changed the key gas remains also H_2.

The proposed technique checked the fault sequences as PD, D1, D2, T1, T2 and T3. This sequence is considered due to its high performance to detect the transformer fault.

Procedure of the proposed algorithm

Based on the gas concentration limits shown in Fig. 2 and summarized in Table 3, the rules to determine the fault types are as in the flowchart depicted in Fig. 3. Step by step process to determine the fault types from the flowchart is as follows:

- The dissolved gases' data set is used as the input data; therefore, the total combustion gas ratios resulting from (2) are calculated.
- In the Step 1 in Fig. 3, the PD limits for each gas concentration are examined that are in the first row in Table 3. If the PD limits are satisfied, then the concentration limits of the other fault

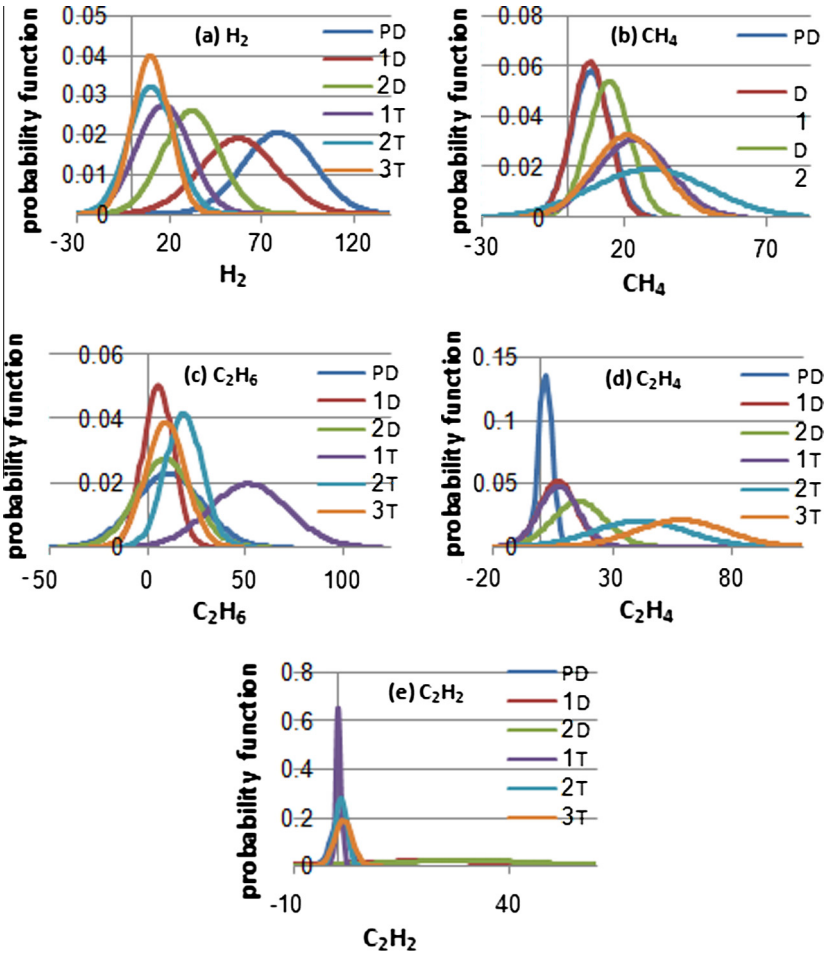


Fig. 4. Continuous probability distribution function for each dissolved gases at different fault type.

types will be checked that either a salient PD fault decision is the indicated or other decision of both PD and the satisfied fault is specified. In other words, if the PD limits are satisfied and if limits of one of the other fault types are satisfied, then the interference between this fault type and PD is concluded.

- When the PD limits are not satisfied, then the step 2 is evaluated that the D1 fault limits (row 2 in Table 3) are checked. If D1 conditions are satisfied, then the limits of the other faults are evaluated after discarding the PD of the first step. Next, if the other faults are not satisfied, then, the fault type is the D1 only. Otherwise, the fault identification is D1 and the other satisfied fault type.
- The processes will be repeated concerning the evaluation of D2 up to T3 (Step 6) and then the UD as depicted in Fig. 3. When the gas concentration limit for T3 is satisfied, then the T3 is the fault, but if not satisfied the fault now can be identified, and then the undetermined fault case is resulted.

Performance evaluation

Some suggestions are taken into account in order to evaluate and calculate the accuracy of the new approach DGA technique and then comparing with the other traditional technique as IEC standard code, Duval triangle and modified Rogers's method. The procedure considered in the evaluation is that the accuracy flag (*AF* in (3)) is 0.5 when interference is taken place between the fault types. On the other hand, if the agreement occurs between the estimated decision and the actual fault, the accuracy flag is 1 however; it will be zero for otherwise (disagreement between the decision and the actual fault type) as in:

$$AF = \begin{cases} 1 & \text{if diagnose} = \text{actual fault} \\ 0 & \text{if diagnose} \neq \text{actual fault} \\ 0.5 & \text{if interference case} \end{cases} \quad (3)$$

The results obtained from a new approach DGA compared to the traditional DGA techniques are illustrated in Tables 4 and 5. Table 4 shows the agreement accuracies as a percentage between the new approach DGA results and the results from traditional DGA techniques (IEC Code Standard, Duval triangle method and modified Rogers's method) according to the fault types. From Table 4, the new approach DGA gives the highest agreement overall accuracy with the actual fault types than that of the other traditional DGA techniques. Some fault types of the other techniques give better accuracy than the new approach DGA, i.e. in case of D1, 60.14% is attained for new approach DGA while it is 72.46% for the same fault with the Duval Triangle method. The reduction in the new approach DGA technique's accuracy in some fault type attributes to the interference that occurs between the fault types. Table 5 shows the agreement accuracy between the new approach DGA and the other DGA techniques according to selected samples of

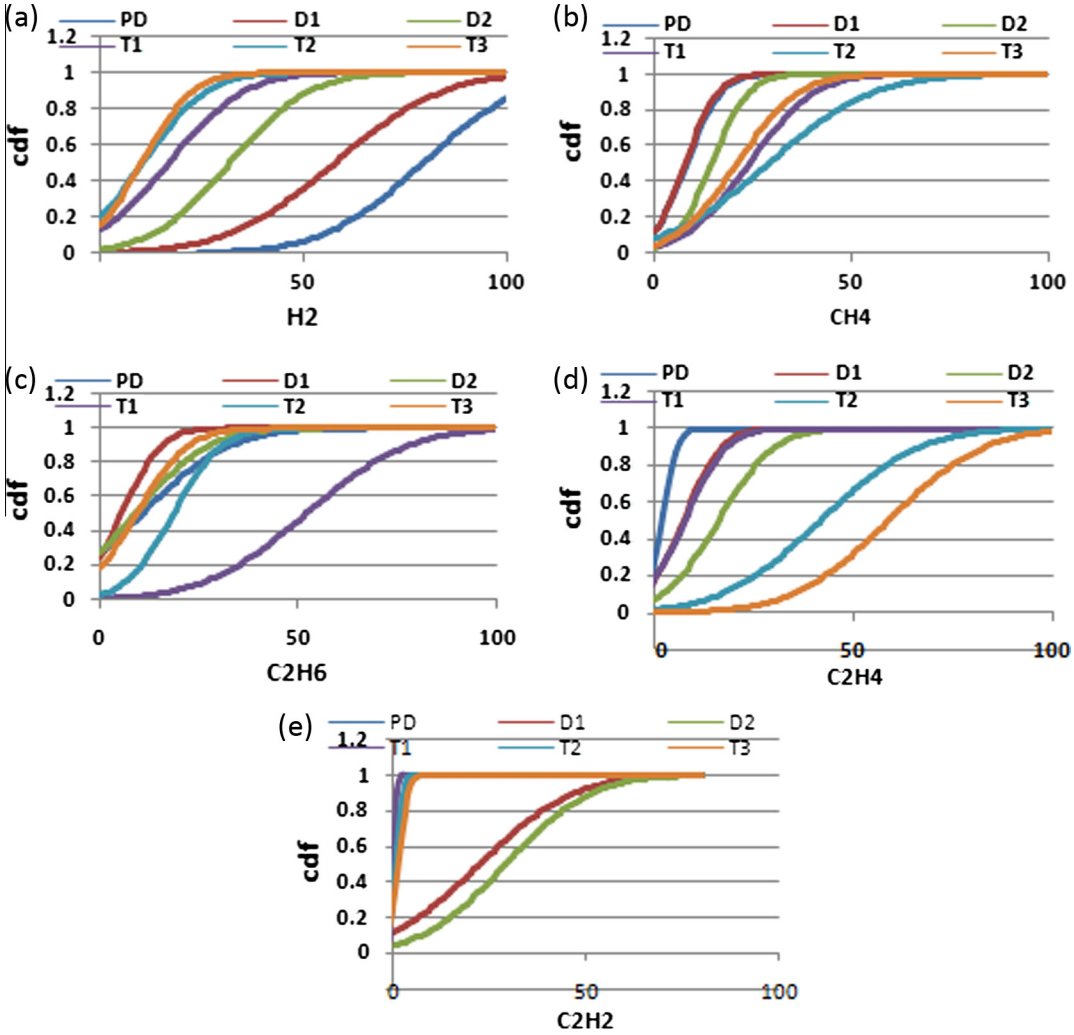


Fig. 5. Cumulative distribution function (CDF) for each dissolved gases at different fault type.

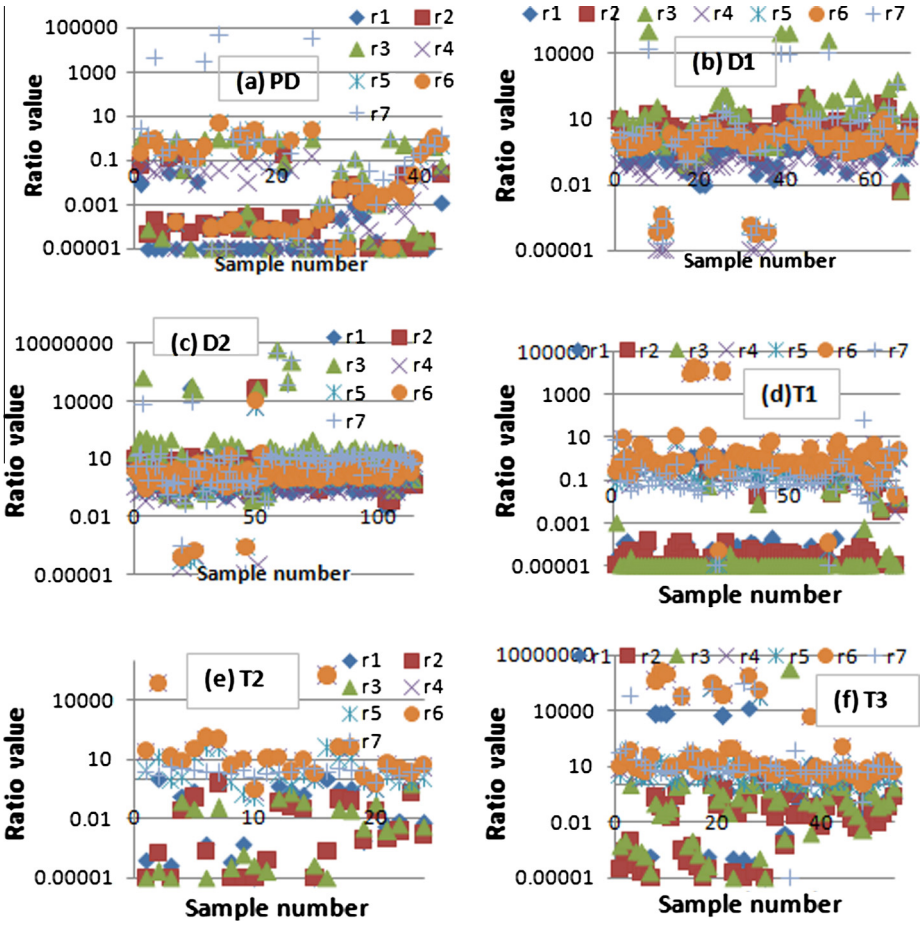


Fig. 6. Gases ratios value according to the fault types.

the literatures data set. As mentioned in Table 4, the overall agreement accuracy for the new approach DGA is the best DGA technique. Although the new approach DGA has a higher accuracy than the other traditional DGA techniques, but a lot of cases have interference between the fault types such as samples illustrated in Table 6. An overlap in many cases between two fault type (PD, D1), (D1, D2), (D2, T1), (T1, T2) and (T2, T3) is observed.

As mentioned before, the proposed algorithm has better performance than the published ones (IEC, Duval, and modified Rogers). However, its performance is 71.5% of successful identification of the fault type. The reason is due to the overlap in the fault features extracted using DGA. In order to illustrate this overlap, the probability distribution function (PDF) that based on the mean and standard deviation of dissolved gases through the samples for each fault type is explained as in Fig. 4. The mean and standard deviation for each of the main five gases are calculated and therefore, the probability distribution function is evaluated for each fault type as in Fig. 4. The continuous probability distribution function of each gas and the interference of each gas with the fault type are investigated.

The probability density function (PDF) and the cumulative distribution function (CDF) are used to express the probability distribution function of the dissolved gases. They explain the overlapping between the fault types as shown in Fig. 4 and verified in Fig. 5. The results in these figures confirm that a great issue of the interference will affect on the percentage accuracy of the DGA interpretation technique. Fig. 4 shows that the means and standard deviations of different key gases for the studied fault cases are overlapped in a presentation of the normal distribution function. The PDF of H_2 as shown in Fig. 4 confirms overlaps between faults (PD, D1), (PD, D2), (D1, D2), (D2, T1, T2, T3) and (T1, T2, T3). This affects on the cumulative distribution function shown in Fig. 5. For example in Fig. 5a, the probability distribution using the Key gas H_2 can support to differentiate between faults PD, D1, and D2 however this gas could contribute to distinguish between T1, T2 and T3 faults due to their overlap. However, Fig. 5b confirms an overlap of different fault types using the gas CH_4. Further confirmation of overlap between the fault detection based on the other key gases is found in Fig. 5c–e. However, such figures of the distribution probabilities provide an evidence to find an algorithm to overcome the overlap features of fault types. Although the normal and cumulative probability distributions in Figs. 4 and 5 succeeded to show and confirm the overlap between the fault type features based on created key gases, they could not contribute to detect the fault type. Further complicated probability-based technique such as Bayesian probability can be used to detect the fault type, and this is going to be evaluated in a work to follow. However, these detected overlap and non-overlap features supported to outline the proposed algorithm. In the next section, an improvement in the new approach DGA technique is made using the ratios between the percentage gas concentrations to separate the interference between the fault features.

Table 7
Modified new approach DGA gas concentration limits and the gases ratios that separate the overlap between fault types.

FT	PD	D1	D2	T1	T2	T3
Ratios	$r_2 \leqslant 0.2$, $r_4 \leqslant 0.16$	$r_1 \leqslant 2$, $r_3 \geqslant 0.1$, $r_4 \leqslant 0.7$	$r_2 \geqslant 0.1$, $r_3 \geqslant 0.14$, $r_4 \leqslant 1.8$	$r_6 \leqslant 5$, $r_7 \leqslant 0.8$	$r_7 \leqslant 4$	None

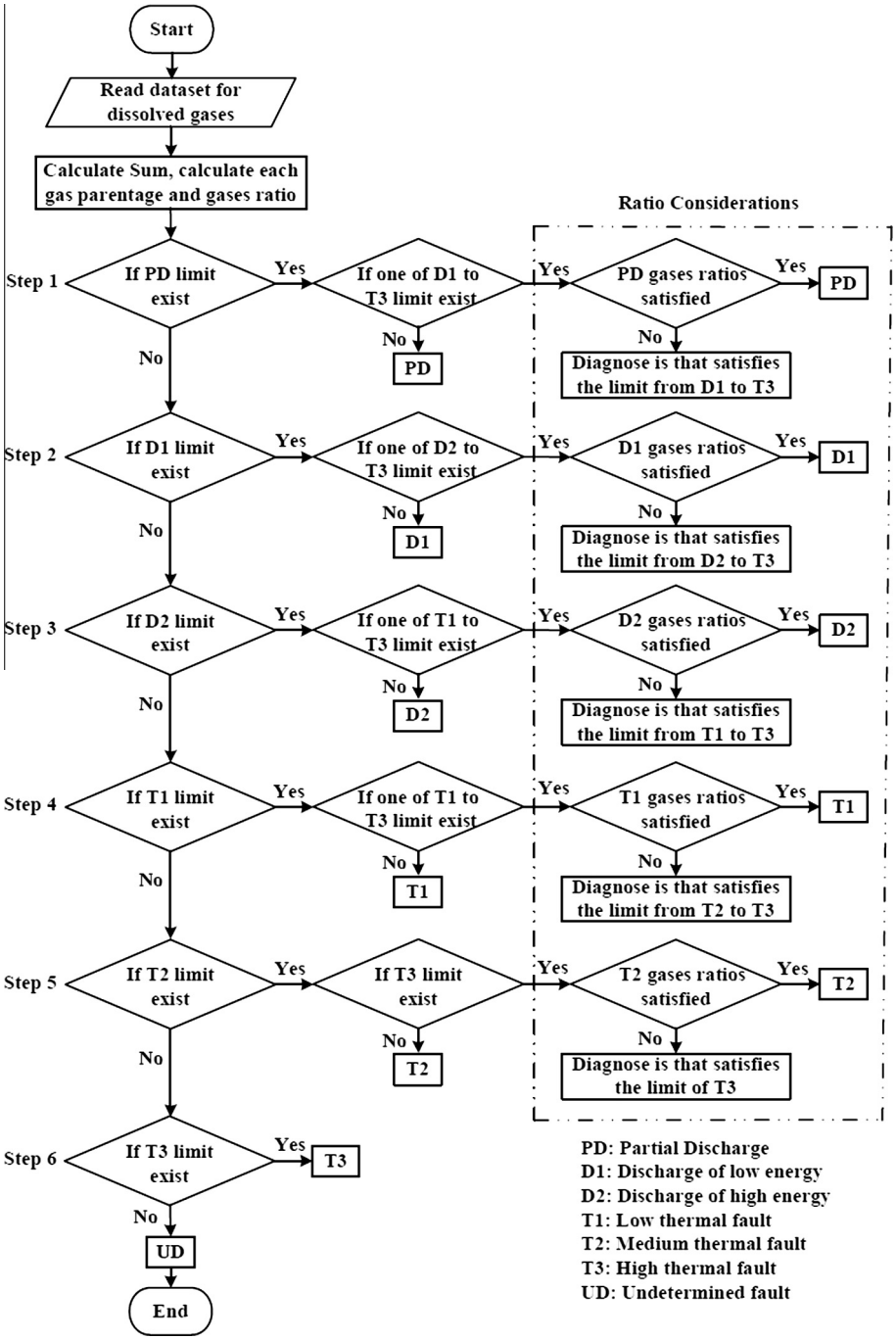


Fig. 7. Flow chart of the new version of new approach DGA technique.

Table 8
Comparison between modified New approach DGA and first version of new approach DGA according to transformer fault types for 386 samples.

Fault type	Sample	New approach DGA (%)	Modified new approach DGA (%)
PD	43	56.97	95.34
D1	69	60.14	76.81
D2	115	88.69	85.215
T1	81	73.23	91.35
T2	24	64.58	62.5
T3	54	75	85.18
Overall	386	71.5	84.71

As aforementioned, the proposed algorithm has better performance than the published ones (IEC, Duval, and modified Rogers). However, its performance is 71.5% of successful identification of the fault type. The reason is due to the overlap in the fault features extracted using DGA. In order to detect this overlap, Fig. 4 shows the probability distribution function (PDF) based on the mean and standard deviation of dissolved gases through the samples for each fault type. The mean and standard deviation for each of the main five gases are calculated and therefore, the probability distribution function is evaluated for each fault type as in Fig. 4. The continuous probability distribution function of each gas and the interference of each gas with the fault type is investigated. Fig. 4 illustrated that a great issue of the interference will affect on the percentage accuracy of the DGA interpretation Technique.

The probability distribution function (PDF) of H_2 as shown in Fig. 4 confirms overlaps between (PD, D1), (PD, D2), (D1, D2), (D2, T1, T2, T3) and (T1, T2, T3). The problem now is how to separate between the fault types. In the next section, an improvement in the new approach DGA technique is made using the ratios between the percentage gas concentrations to separate the interference between the fault features.

Ratio based improving the proposed DGA interpretation technique

It is very important to solve the problem of overlaps between the fault types with the same limit as in Fig. 1. Therefore, the *CGP* with each fault type as well as some gas ratios was developed. Details about the gas ratios are explained as in (4).

$$\begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \\ r_6 \\ r_7 \end{pmatrix} = \begin{pmatrix} C_2H_2/H_2 \\ C_2H_2/CH_4 \\ C_2H_2/C_2H_6 \\ C_2H_4/H_2 \\ C_2H_4/CH_4 \\ (C_2H_4/H_2)+(C_2H_4/CH_4) \\ C_2H_4/C_2H_6 \end{pmatrix} \quad (4)$$

Some of these gases ratios are assumed based on the gas concentration in order to separate the overlaps among the faults. These ratios are defined based on the variation of its value through the fault types. Fig. 6 shows the ratios' value varies through the sample according to fault types. The same data set is applied to the second version of approach DGA that is used to prevent the overlaps between the fault types and enhance the accuracy of the first version of the new approach DGA technique. The suggested ratios r_2 and r_4 as in Table 7 distinguish the PD fault. On the other hand, r_1, r_3 and r_4 limits point out the low discharge energy arcing (D1). Further ratio limit considerations are summarized in Table 7 that they can contribute to distinguishing each fault type. The ratio limits are investigated and proposed as in Table 7 by empirical study to satisfy the best performance of the technique to prevent the overlapping between the fault types.

A modified flowchart as in Fig. 7 is built with proposed gas ratios to illustrate the step-by-step process to diagnose the transformer fault. It is the same procedure proposed in Fig. 3, however, added ratio conditions are considered for the six steps. For example, the PD limits for each gas concentration are checked. If the PD limits are satisfied, then the concentration limits of the other fault types will be checked. If the limit of the PD is not satisfied, hence, the D1 fault will be checked. Then, if the PD limits are satisfied and if limits of one of the other fault types are satisfied, afterwards, the interference between this fault type and PD is concluded. To identify the fault type, then the gas ratios that can separate between the overlaps faults will be checked for the PD fault as in Table 7. If these gas ratios are satisfied hence, the fault will be PD and if not satisfied the fault will be the faults that limits are satisfied. On the other hand, if the limit of the other faults is not satisfied, then, the fault type is the PD. The processes will be repeated, if it is needed, concerning the consequences steps to reach to the T3 or UD.

Table 9
Comparison between modified new approach DGA and first version of new approach DGA according to the source of dataset for 386 samples.

Reference	Samples	New approach DGA (%)	Modified new approach DGA (%)
Lab	240	75.38	92.91
Ref. [10]	8	43.75	37.55
Ref. [15]	5	70	100
Ref. [17]	99	68.18	71.71
Ref. [18]	19	55.26	68.42
Ref. [19]	6	58.33	83.33
Ref. [20]	9	61.11	77.77
Overall	386	71.5	84.71

The diagnostic processes for the second version of new approach DGA interpretation technique will be explained. Table 8 illustrates the comparison between the modified new approach DGA techniques and the first version of the new approach DGA according to the fault types. The improvement takes place of the agreement percentage between the new approach DGA and modified one of the fault types. A significant increase in the accuracy percentage is attained especially for the electrical faults (PD, D1, T1, and T3) but a slight reduction on the D2 and T2 is observed. However, the overall accuracy is improved. Fig. 8 illustrates the comparison between the modified new approach DGA techniques and the common DGA interpretation methods such as IEC code, Duval Triangle method and Rogers' method based on four ratios according to the fault types. There is a large difference of the agreed percentage between the new approach DGA and the common interpretation techniques for the fault types, especially for the electrical faults (PD, D1 and D2) and T1 for the thermal faults. In case of T2 and T3 thermal faults the agreed percentage for the proposed DGA is very close according to the others DGA techniques.

Table 9 illustrates the comparison between the modified new approach DGA techniques and the first version of the proposed DGA according to the source of the dataset. A significant increase in the accuracy percentage is observed for all data sources, except dataset for Ref. [10]. The overall accuracy is increased from 71% to 84.71%; therefore, a significant improvement will be made using the modified new approach DGA.

Fig. 9 explains the agreement accuracy percentage between the new approach DGA and the common interpretation techniques

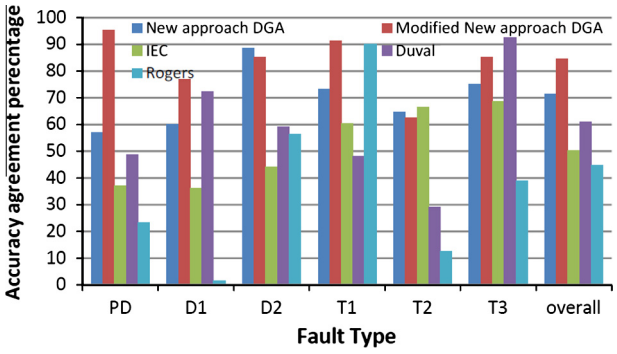


Fig. 8. Agreement percentage according to actual faults for DGA interpretation techniques (386 samples).

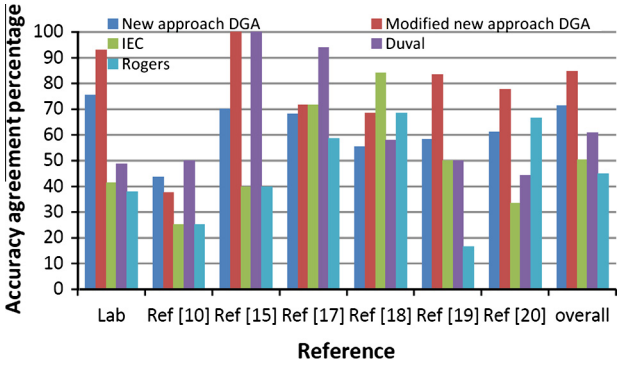


Fig. 9. Agreement accuracy with actual faults for DGA interpretation techniques (386 samples).

Table 10
Solving the interference problem using the Improved DGA interpretation technique for the same dataset in Table 6.

Gases concentration (ppm)					Ref.	Actual	Modified new approach DGA	New approach DGA	Modified DGA agree
H_2	CH_4	C_2H_6	C_2H_4	C_2H_2					
117	17	1	3	1	Lab	PD	PD	PD, D1	1
32,930	2397	157	0.001	0.001	[17]	PD	PD	PD, D1	1
78	20	11	13	28	[17]	D1	D1	D1, D2	1
1230	163	27	233	692	[17]	D1	D1	D1, D2	1
8200	3790	250	4620	277	[17]	D2	D2	D2, T1	1
13	3	1	3	6	[17]	D2	D1	D1, D2	0
130	140	2	120	0.001	[15]	T1	T1	D2, T1	1
78	66	283	2.6	0.001	[15]	T1	T1	D2, T1	1
30.4	117	44.2	138	0.1	[10]	T2	T2	T2, T3	1
27	90	42	63	0.2	[18]	T2	T1	T1, T2	0
1100	1600	221	2010	26	[17]	T3	T3	T2, T3	1
290	966	299	1810	57	[17]	T3	T3	T2, T3	1

according to the source of the sample data set. From the results shown in Fig. 9, there is a large difference between the new approach DGA and Duval triangle method that is the best diagnostic DGA technique. It is seen from Fig. 9 that the overall accuracy of the new approach DGA is 84.71% compared with 60.88% of the Duval Triangle interpretation technique. This explains the high dependability of the New approach DGA in the transformer fault diagnostic. Table 10 explains the modification that developed by the modified version of the new approach DGA to separate the interference between the fault types.

Conclusions

A new approach DGA Technique has been developed based on the gas concentrations according to five main dissolved gases in the oil transformers, H_2, CH_4, C_2H_6, C_2H_4 and C_2H_2. Firstly, the gas concentration limits have been evaluated in order to define each transformer fault type. The new approach DGA technique provided higher agreement accuracy with the actual fault than the traditional common interpretation DGA techniques such as IEC Code, Rogers' method and Duval triangle method. However, a lot of cases suffer from the interference between the transformer fault types. Study the continuous statistical function of gases' distribution at the transformer fault types has referred to a great interference between the transformer fault types. The proposed DGA interpretation technique has been modified by taking into account some gas ratios that are presented based on an empirical study to overcome the problem of interference using the gas concentration ratios. The overall accuracy of the new approach DGA technique is enhanced to 84.71% compared with 50.26%, 60.88% and 44.82% of the IEC, Duval Triangle and Rogers' methods respectively. The accuracy of the modified new approach maybe give low accuracy comparing to the artificial intelligent DGA techniques due to the large number of test samples that take into account for. The results provided evidence the efficacy of the proposed DGA technique for transformer fault diagnoses comparing with the traditional based ratio methods.

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